NATIONAL INSTITUTE OF TECHNOLOGY, TIRUCHIRAPPALLI-620015

CSPE64 - Data Analytics
Mini Project Titles

Sl.No	Topic
1.	**Rule-Based Named Entity Recognition for Indian Languages** **Problem Statement:** Design a system to identify person, location, and organization names in Indian language text using linguistic rules and gazetteers. **Input:** Raw sentences in an Indian language (Tamil/Hindi/Telugu). **Output:** Annotated text with labeled named entities. **Resources**: 1. Indic NLP Library: https://indicnlp.ai4bharat.org/ 2. ICON NLP Tools: https://ltrc.iiit.ac.in/icon2015/ 3. https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0179488 4. https://researcharchive.lincoln.ac.nz/entities/publication/480d78b5-9941-4e12-9959-ca745c71f182 5. https://aclanthology.org/W10-2419.pdf 6. https://aclanthology.org/D10-1098.pdf 7. https://d1wqtxts1xzle7.cloudfront.net/90248264/comlex00-libre.pdf?1661442029=&response-content-disposition=inline%3B+filename%3DRule_based_named_entity_recognition_for.pdf&Expires=1772775127&Signature=TsLV3-TaF0EkjhonYnAzjL~m6Rakceq3lb8qGTPSh6imFj4JzCAZUbKm9TfDbhCFsVnUmir~SFMQ86Rmahi3yhu-HglFQJmquMSMwwpxUgwWdBGHgLrNDpcSrvfc6Ul-mleRT6TcV~fC-kGxwNSBTqMj9dS4H9tXtnuLGbti9PmvQ~A-4eZUDd8-HpA7ba-7qRFtjOIULQtj17BjKEgRFr9LBxwj0ItOzN1tGCEBzdULQyMLsU9CK6O1-BF4AKEc2xz-C35W7mFsSPjQH~Wino~bsQ-ZL~zdAwuTGb4IHnBPG35MqjdxHOv8dGc2eLERfJ~ltOXjPmYV43bultI-Xw__&Key-Pair-Id=APKAJLOHF5GGSLRBV4ZA

	8. https://www.isca-archive.org/icslp_2000/kim00_icslp.pdf 9. https://ieeexplore.ieee.org/abstract/document/5561392 10. https://dl.acm.org/doi/abs/10.1145/2090176.2090178
2.	**Machine Learning-Based Biomedical Nested NER Using Feature Engineering** **Problem Statement:** Develop an NER system using traditional ML classifiers and handcrafted linguistic features for biomedical nested NER. **Input:** Tokenized and POS-tagged text corpus. **Output:** Entity-tagged sequence output. **References:** 1. CoNLL-2003 Dataset: https://www.clips.uantwerpen.be/conll2003/ner/ 2. CRF++ Toolkit: https://taku910.github.io/crfpp/ 3. https://aclanthology.org/P19-1585.pdf 4. https://aclanthology.org/D09-1015.pdf 5. https://aclanthology.org/N18-1131.pdf 6. https://aclanthology.org/2022.coling-1.184.pdf 7. https://aclanthology.org/2024.emnlp-main.492.pdf 8. https://aclanthology.org/P19-1527.pdf 9. https://aclanthology.org/W06-3318.pdf

3.	**Knowledge Graph Construction through LLM from Indian News Articles** **Problem Statement:** Extract entities and relationships from Indian news articles to build a structured knowledge graph. **Input:** Preprocessed news text. **Output:** Entity–relation triples stored as a graph. **References:** 1. DBpedia: https://www.dbpedia.org/ 2. GDELT Project: https://www.gdeltproject.org/ 3. https://ceur-ws.org/Vol-3753/paper6.pdf 4. https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=11316475 5. https://repositum.tuwien.at/bitstream/20.500.12708/220444/1/Wilberg%20Felix%20-%202025%20-%20Large%20Language%20Model%20based%20Knowledge%20Creation%20Verified%20by...pdf 6. https://www.mdpi.com/2504-4990/7/2/38 7. https://aclanthology.org/2025.nllp-1.11.pdf 8. https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=11375209 9. https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10825937 10. https://arxiv.org/pdf/2509.09272?

4.	**Ontology-Based Question Answering System Biomedical Text** **Problem Statement:** Create a QA system that answers fact-based questions using a domain ontology. **Input:** Natural language question. **Output:** Exact answer retrieved from ontology. **References:** 1. Apache Jena: https://jena.apache.org/ 2. OWL Ontology Language: https://www.w3.org/OWL/ 3. https://link.springer.com/content/pdf/10.1186/s13326-015-0029-x.pdf 4. https://www.sciencedirect.com/science/article/pii/S0169023X06000929 5. https://www.mdpi.com/2078-2489/12/5/200 6. https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9802991 7. https://link.springer.com/content/pdf/10.1186/s12859-018-2339-3.pdf 8. https://aclanthology.org/W18-5310.pdf 9. https://ceur-ws.org/Vol-3603/Paper9.pdf 10. https://link.springer.com/content/pdf/10.1007/s40593-018-00172-w.pdf
5.	**Knowledge Graph-Based Search through RAG (Academic/News Papers etc.)** **Problem Statement:** Develop a search system that retrieves academic papers using entity-based indexing. **Input:** User query with keywords. **Output:** Ranked list of relevant papers. **References:** 1. Semantic Scholar Open Research Corpus: https://www.semanticscholar.org/ 2. OpenAlex: https://openalex.org/ 3. **https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=11035854** 4. https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10871140

	5. https://arxiv.org/pdf/2410.20724 6. https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10892619 7. https://arxiv.org/pdf/2508.08344? 8. https://arxiv.org/pdf/2508.05318? 9. https://watermark02.silverchair.com/ocaf059.pdf?token=AQECAHi208BE49Ooan9kkhW_Ercy7Dm3ZL_9Cf3qfKAc485ysgAAA4EwggN9BgkqhkiG9w0BBwagggNuMIIDagIBADCCA2MGCSqGSIb3DQEHATAeBglghkgBZQMEAS4wEQQMcnf89MI9ZOUgSjwcAgEQgIIDNGCjsoh7I6woCsPaNIQOERBgfrntxKTbAt-eF61uwEOU2l9d79SEqSOx3Pfot-e1-jvO4XbDOJzFCjb4bdkD1UuIV5Yw44ZmABl8qg7LupJEdzddEzG2rSY5nKiNbrdwk2ZoiThSVhl2f_SnNhM4pXW-1TRAS7Oh6i1e0e8U8caQ2r8SosFW61VEX52wfcv181_IyPdUnDOxFEWP0Fiq39xiMTedV-6ZI6nT8mz46yU0SiJ3RfY7vC5MUw5LzyNBJjSzIqxo3AA4K_fLyppnTbjv7MDc-7FtygSWrS_0DzngCJ6DzIXsi5SaeRqHIjpNOPmdnpzZ2bZj7VUCI1oYXJntYzERDlQIdldQRTo4sx_vR3uv1M3XmVk51LqQ_cxopydRKMqqmyVha1oZl7VsNRzx-KrfSCUZPbPOkFxDe7xFKVhxa6ef-AS6t50GIp8GAXy_jzi0C1AuQYNcLQARq8AFML5e-IlJnOODgTaHq1JQX1f-FaLN3lz02X_gsrqsVYdUcNnqJvmsy77vrepUa8oKdUJP7tpLgq3NFBBnBXaFz6xg-py3Uxil2nkpYtGJrhvOcXjUN-D4DYTwv4WyQCDmRVEfJ4dfnvfBz_IkObo8GnWiaIYfkFyO0oDBUzXsz-LOdXKIEpyO2Ekn6fnFIpzFyajzyaGUrHVOsbSB7r38mp8q6jXhJOQxwksZQccvZ1wXBkDeOrhm0LxL7ubryywqsRA1T0V_gfyyjqE5xMTZGNOwAwg12_RKf2PxZZabvgxG2qKsnw-1h0Gq98a1SudfLfyQPWEHJYgfqseDrtmK2ww5aQMjxlfE8OeeCyCKNF-wmDCXeOj5_S-EUNOQsKzlDgdX-Ekg_XeS_X-pWtfg-k3WGUFCoWXBfQvvMzsMzz7V_gLIIpJGvzZGRjkg-d9_xxp7tGoHJFAjOzg3x_yUS0hOjAQjUqfTKEHEgL-v3-ytUKaKD0d-2X4PvibKM-7wNz_o_bkWHkOA9t4qIhMntED1iQJn_gn915Kt7UKoLJ12xv0bjyA2Pnym78ri16dCefl7bP--P1vJs7v2FqJ0-cI6oxFdlA_7ZZSVuDefDb_o4eA
6.	**Visual Question Answering Using Object Attributes** **Problem Statement:** Answer questions about images using object annotations and rule-based reasoning. **Input:** Image + question. **Output:** Text-based answer. **References:** 1. CLEVR Dataset: https://cs.stanford.edu/people/jcjohns/clevr/ 2. Visual Genome (metadata): https://visualgenome.org/ 3. https://dl.acm.org/doi/pdf/10.1145/3219819.3220036 4. https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=7934440 5. https://openaccess.thecvf.com/content_iccv_2015/papers/Antol_VQA_Visual_Question_ICCV_2015_paper.pdf 6. https://openaccess.thecvf.com/content_ICCV_2017/papers/Kafle_An_Analysis_of_ICCV_2017_paper.pdf 7. https://arxiv.org/pdf/2008.11976 8. https://openaccess.thecvf.com/content_cvpr_2017/papers/Ramakrishnan_An_Empirical_Evaluation_CVPR_2017_paper.pdf

	9. https://www.researchgate.net/profile/Farhad-Shakerin/publication/338639933_AQuA_ASP-Based_Visual_Question_Answering/links/5e2795a9299bf152167318c9/AQuA-ASP-Based-Visual-Question-Answering.pdf 10. https://link.springer.com/content/pdf/10.1007/s10462-020-09832-7.pdf 11. https://aclanthology.org/2025.paclic-1.36.pdf
7	**Multilingual NER for Indian Government Documents** **Problem Statement:** Identify entities in multilingual government documents using rule-based and statistical methods. **Input:** Documents in multiple Indian languages. **Output:** Structured entity lists. **References:** 1. India Open Government Data: https://data.gov.in/ 2. TDIL Resources: https://tdil-dc.in/ 3. https://aclanthology.org/2024.naacl-srw.9/ 4. https://arxiv.org/pdf/2405.04829 5. https://github.com/RXO95/ner-indian-languages 6. https://link.springer.com/article/10.1007/s42452-020-2983-x 7. https://dl.acm.org/doi/10.1145/3590837.3590892

8	**Intent Detection through LLM and RAG for Information-Seeking Questions** **Problem Statement:** Identify the underlying user intent in information-seeking questions to improve answer retrieval efficiency using classical machine learning methods. **Input:** User questions expressed in natural language. **Output:** Predicted intent label (e.g., definition, procedure, comparison, factual lookup). **References:** 1. TREC QA Dataset: https://trec.nist.gov/data/qamain.html 2. Scikit-learn Classical ML Models: https://scikit-learn.org/stable/supervised_learning.html 3. https://aclanthology.org/2020.nlp4convai-1.5.pdf 4. https://dl.acm.org/doi/pdf/10.1145/3547138 5. https://dl.acm.org/doi/pdf/10.1145/1507509.1507510 6. https://aclanthology.org/2024.emnlp-industry.114.pdf 7. https://aclanthology.org/2020.aacl-main.35.pdf 8. https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=7846297 9. https://aclanthology.org/D18-1348.pdf
9	**Entity Linking in Indian Language Text for Wiki data** **Problem Statement:** Link extracted entities to corresponding knowledge base entries. **Input:** Named entities. **Output:** Knowledge base identifiers. **References:** 1. Wikidata: https://www.wikidata.org/ 2. Indic NLP Library: https://indicnlp.ai4bharat.org/ 3. https://aclanthology.org/2023.ranlp-1.2.pdf 4. https://aclanthology.org/D17-1284.pdf

	5. https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10817806 6. https://www.sciencedirect.com/science/article/pii/S1877050916311590/pdf?md5=b62268346b7fe9e618a7384598d498a5&pid=1-s2.0-S1877050916311590-main.pdf&_valck=1 7. https://aclanthology.org/W12-3902.pdf 8. https://dl.acm.org/doi/pdf/10.1145/3015467 9. https://dl.acm.org/doi/pdf/10.1145/3795134 10. https://arxiv.org/pdf/2112.01989 11. https://journals.sagepub.com/doi/pdf/10.3233/SW-212865
10	**Rule-Based Visual Scene Understanding** **Problem Statement:** Describe object relationships in images using spatial reasoning rules. **Input:** Image object metadata. **Output:** Scene relationship descriptions. **References:** 1. Pascal VOC Dataset: http://host.robots.ox.ac.uk/pascal/VOC/ 2. OpenCV Documentation: https://docs.opencv.org/ 3. https://www.sciencedirect.com/science/article/pii/S0262885614001395 4. https://ieeexplore.ieee.org/iel7/6895053/6906581/06907238.pdf 5. https://www.sciencedirect.com/science/article/pii/S1077314217302291 6. https://ieeexplore.ieee.org/iel1/34/216/00003867.pdf 7. https://www.mdpi.com/2220-9964/10/1/32

11.	**Slot Filling for Indian Language Text books QA through LLM and RAG** **Problem Statement:** Extract structured slots from user questions. **Input:** User question text. **Output:** Slot–value pairs. **References:** 1. SNIPS Dataset: https://github.com/snipsco/nlu-benchmark 2. Stanford NLP (classic tools): https://nlp.stanford.edu/software/ 3. https://papers.ssrn.com/sol3/Delivery.cfm?abstractid=4761601 4. https://www.sciencedirect.com/science/article/pii/S0031320324010835 5. https://ieeexplore.ieee.org/iel8/10816706/10816707/10817016.pdf 6. https://ieeexplore.ieee.org/iel8/10716817/10716818/10717078.pdf 7. https://arxiv.org/pdf/1902.10909 8. https://aclanthology.org/N18-2118.pdf 9. https://aclanthology.org/D17-1004.pdf 10. https://ieeexplore.ieee.org/iel7/6570655/6633080/06998838.pdf 11. https://arxiv.org/pdf/1609.01454

12.	**Knowledge Graph-Based Recommendation System** **Problem Statement:** Recommend items using graph traversal and similarity metrics. **Input:** User preferences and KG. **Output:** Top-N recommendations. **References:** 1. MovieLens Dataset: https://grouplens.org/datasets/movielens/ 2. RDF Primer: https://www.w3.org/TR/rdf11-primer/ 3. https://ieeexplore.ieee.org/iel7/10450150/10449968/10450693.pdf 4. https://www.sciencedirect.com/science/article/pii/S2090447923001521 5. https://www.sciencedirect.com/science/article/pii/S138650562400217X 6. https://ieeexplore.ieee.org/iel7/69/4358933/10058002.pdf 7. https://www.mdpi.com/2076-3417/12/19/9712 8. https://www.sciencedirect.com/science/article/pii/S0020025522001967 9. https://link.springer.com/content/pdf/10.1007/s00521-021-06573-8.pdf 10. https://ieeexplore.ieee.org/iel7/2945/4359476/09552844.pdf
13.	**NER Error Analysis in Low-Resource Indian Languages using Vector Search in RAG** **Problem Statement:** Analyze errors in NER outputs for low-resource languages. **Input:** Gold vs predicted tags. **Output:** Error statistics. **References:** 1. ILCI Corpus: https://ltrc.iiit.ac.in/ilci/ 2. BIO Tagging Scheme: https://www.clips.uantwerpen.be/conll2000/chunking/ 3. https://norma.ncirl.ie/5178/1/pavankulkarni.pdf

	4. https://www.cambridge.org/core/services/aop-cambridge-core/content/view/7D3DA31DB6C01B13C6B1F698D4495951/S2977042424000335a.pdf/div-class-title-natural-language-processing-applications-for-low-resource-languages-div.pdf 5. https://arxiv.org/pdf/2601.02065 6. https://ieeexplore.ieee.org/iel8/11182305/11182297/11182476.pdf 7. https://ieeexplore.ieee.org/iel8/10887540/10887541/10889964.pdf 8. https://aclanthology.org/2025.wsc-csdh.11.pdf 9. https://papers.ssrn.com/sol3/Delivery.cfm?abstractid=5797544 10. https://link.springer.com/content/pdf/10.1007/s41666-025-00190-z.pdf
14.	**Factoid Question Answering from Wikipedia** **Problem Statement:** Answer factual questions using entity matching and keyword retrieval. **Input:** User question. **Output:** Single factual answer. **References:** 1. Wikipedia Dumps: https://dumps.wikimedia.org/ 2. Lucene Search Engine: https://lucene.apache.org/ 3. https://www.sciencedirect.com/science/article/pii/S0306457314000351 4. https://aclanthology.org/D14-1070.pdf 5. https://dl.acm.org/doi/pdf/10.1145/3018661.3018704 6. https://ieeexplore.ieee.org/iel7/7911113/7917923/07917964.pdf 7. https://aclanthology.org/N19-1027.pdf 8. https://www.academia.edu/download/81927542/4313ijwest03.pdf 9. https://dl.acm.org/doi/pdf/10.1145/3209978.3210118 10. https://link.springer.com/content/pdf/10.1007/s10844-021-00655-8.pdf

15.	**Temporal Information Extraction in News Articles using LLMs** **Problem Statement:** Extract and normalize date/time expressions from text. **Input:** News articles. **Output:** Normalized temporal entities. **References:** 1. TimeML: https://timeml.org/ 2. HeidelTime: https://github.com/HeidelTime/heideltime 3. https://dl.acm.org/doi/pdf/10.1145/3674127.3674137 4. https://ieeexplore.ieee.org/iel8/10649807/10649898/10650983.pdf 5. https://arxiv.org/pdf/2602.12537 6. https://aclanthology.org/2023.findings-emnlp.274.pdf 7. https://ieeexplore.ieee.org/iel8/11139126/11139366/11140911.pdf 8. https://aclanthology.org/2024.cl4health-1.18.pdf 9. https://dl.acm.org/doi/pdf/10.1145/3626772.3657899 10. https://aclanthology.org/2024.acl-long.187.pdf
16	**Consistency Validation of Healthcare Knowledge Graphs** **Problem Statement:** Validate a healthcare knowledge graph to detect missing, contradictory, or logically inconsistent relationships between diseases, symptoms, and treatments in order to support reliable clinical decision support systems. **Input:** Healthcare knowledge graph triples (disease–hasSymptom–symptom, disease–treatedBy–medicine). **Output:** List of detected inconsistencies and missing relationships. **References:** 1. SHACL Specification: https://www.w3.org/TR/shacl/ 2. Protégé Ontology Editor: https://protege.stanford.edu/

	3. https://link.springer.com/content/pdf/10.1186/s40537-023-00774-9.pdf 4. https://www.nature.com/articles/s41598-017-16674-x.pdf 5. https://arxiv.org/pdf/2005.01389 6. https://www.sciencedirect.com/science/article/pii/S0957417419308371 7. https://link.springer.com/content/pdf/10.1186/s12911-023-02409-8.pdf 8. https://dl.acm.org/doi/pdf/10.1145/3773278 9. https://arxiv.org/pdf/2305.12788 10. https://onlinelibrary.wiley.com/doi/pdf/10.1155/2017/2858423
17	**Cross-Lingual Place Name Resolution for Indian Government Portals** **Problem Statement:** Resolve and align place names appearing in different Indian languages across government databases to ensure unified access to public services using lexical similarity and dictionary-based matching techniques. **Input:** Lists of location names in two Indian languages (e.g., Hindi–Tamil, Telugu–English). **Output:** Mapped equivalent place-name pairs across languages. **References:** 1. IndoWordNet: https://www.cfilt.iitb.ac.in/indowordnet/ 2. Levenshtein Distance: https://en.wikipedia.org/wiki/Levenshtein_distance 3. Government of India Open Data Portal: https://data.gov.in/ 4. https://dl.acm.org/doi/pdf/10.1145/1390334.1390372 5. https://aclanthology.org/P17-1178.pdf 6. https://aclanthology.org/C98-1036.pdf 7. https://aclanthology.org/W03-1508.pdf 8. https://ieeexplore.ieee.org/iel5/4768845/4777819/04777895.pdf 9. https://dl.acm.org/doi/pdf/10.1145/2533888.2533943 10. https://aclanthology.org/N19-1383.pdf

18	**Structured QA and RAG for Indian Railway Information** **Problem Statement:** Answer railway-related queries using structured databases and rules. **Input:** User question. **Output:** Accurate factual response. **References:** 1. Indian Railways APIs: https://www.data.gov.in/ 2. SQL Tutorial: https://www.w3schools.com/sql/ 3. https://ieeexplore.ieee.org/iel8/11080685/11080705/11080860.pdf 4. https://www.mdpi.com/2220-9964/15/2/78 5. https://aclanthology.org/2025.realm-1.17.pdf 6. https://link.springer.com/content/pdf/10.1007/s10791-025-09823-8.pdf 7. https://ieeexplore.ieee.org/iel8/11210849/11210897/11211435.pdf 8. https://ieeexplore.ieee.org/iel8/11216651/11216431/11216705.pdf 9. https://ieeexplore.ieee.org/iel8/6287639/6514899/11244909.pdf 10. https://link.springer.com/content/pdf/10.1007/s10515-025-00587-5.pdf
19	**LLM and RAG Based Riddle Classification in Indian Languages** **Problem Statement:** Automatically classify Indian language riddles into categories such as logic-based, mathematical, wordplay, or cultural riddles using traditional machine learning techniques. **Input:** Text riddles in Indian languages (Tamil, Hindi, Telugu, Malayalam). **Output:** Predicted riddle category label. **References:** 1. Indic NLP Library: https://indicnlp.ai4bharat.org/

	2. WEKA Machine Learning Toolkit: https:/7/www.cs.waikato.ac.nz/ml/weka/ 3. https://aclanthology.org/2025.paclic-1.36/ 4. https://arxiv.org/pdf/2511.00960 5. https://www.jstor.org/stable/pdf/2798727.pdf 6. http://www.vestnik-philology.mgu.od.ua/archive/v43/part_3/Filologi43_3.pdf#page=167 7. https://www.jstor.org/stable/pdf/1177505.pdf 8. https://kb.osu.edu/bitstreams/8c7813e9-ccde-5da7-bdc1-64a68e7221f4/download 9. https://languageinindia.com/aug2006/riddlesmalayalam.pdf 10. https://www.jnu.ac.in/sites/default/files/crs/critic_Issue_No_21.pdf#page=229
20	**Puzzle Solving System for Indian Languages using AI** **Problem Statement:** Automatically solve text-based puzzles and riddles in Indian languages by identifying key clues and mapping them to correct solutions using linguistic analysis and LLMs, RAG, and traditional machine learning techniques. **Input:** Puzzle or riddle text written in an Indian language. **Output:** Predicted solution or answer to the puzzle. **References:** 1. Indic NLP Library: https://indicnlp.ai4bharat.org/ 2. Scikit-learn (Classical ML Models): https://scikit-learn.org/stable/supervised_learning.html 3. NLTK Documentation: https://www.nltk.org/ 4. https://www.academia.edu/download/124862858/Dept_of_Linguistics_CBW_01_AI_and_Indian_Languages_Prof_Achuthsankar_Report.pdf 5. https://link.springer.com/content/pdf/10.1186/s42269-024-01274-3.pdf 6. https://aclanthology.org/2025.findings-acl.155.pdf 7. https://ieeexplore.ieee.org/iel8/11293843/11293984/11295638.pdf 8. https://www.worldscientific.com/doi/pdf/10.1142/S0218001495000316 9. https://aclanthology.org/2025.naacl-long.104.pdf 10. https://aclanthology.org/2023.arabicnlp-1.23.pdf
20	**Decoding Inner Speech for Direct Communication for Indian Languages** **Problem Statement:** Overcoming the low signal to noise ratio and inter subject variability of non-invasive EEG to decode silent, internally

	articulated words into text. **Input:** Multi-channel raw EEG signals (focusing on frontal and temporal lobes) recorded during "imagined speech" tasks. **Output:** Predicted phonemes, discrete words, or semantic word embeddings. **Resources:** 1. https://link.springer.com/article/10.1007/s11517-025-03478-9 2. https://www.sciencedirect.com/science/article/abs/pii/S1568494625008749
22	**Context Aware Dropout Predictor for Remote Learning** **Problem Statement** : Current online education platforms fail to provide real-time insight into student cognitive states, causing teachers to identify confusion or disengagement only after a student has already fallen behind. This project aims to analyze behavioral metadata (time spent on a page, frequency of pausing videos, rate of submitting homework etc) to predict dropout risk weeks in advance, enabling proactive instructor intervention. **Input:** Student interaction in a virtual learning environment **Output:** 1. Risk Score: A numerical probability indicating the likelihood of a student failing or dropping out within the next few weeks. 2. Engagement Metrics: Visual dashboards highlighting specific behavioral changes 3. Actionable Interventions: AI-generated recommendations for the teacher (send a personalized encouraging email, suggest a review session on topic Y). **Reference:** 1. https://analyse.kmi.open.ac.uk/open-dataset#data 2. https://www.sciencedirect.com/science/article/pii/S2666920X22000212
23	**Real time Neuroplasticity Feedback for Mental Health** **Problem Statement:** Using unsupervised deep learning to identify real time EEG biomarkers of mental health trajectories (e.g., anxiety or depression) to provide immediate therapeutic neurofeedback. **Input:** Continuous streaming EEG data **Output:** Real-time "Mental Health Score" or generative visual/auditory feedback that guides the user toward healthier neural states. **Resources:**

	1. https://www.mdpi.com/2813-9844/7/3/61
24	**EEG to Image: Visualizing the "Mind's Eye"** **Problem Statement:** Mapping EEG spectral features to the latent space of a Diffusion Model to reconstruct visual imagery from the "covert" (hidden) brain activity during sleep or intense visualization. **Input :** Raw EEG signals aligned with image-stimuli data **Output:** A generative image/reconstruction of the user's internal visual state.1111111 **Resources:** 1. https://www.nature.com/articles/s41598-025-21635-w
25	**ADHD detection using Machine Learning** **Problem statement:** Attention Deficit Hyperactivity Disorder (ADHD) diagnosis in children is largely based on subjective behavioral assessments, which can lead to delayed or inconsistent identification. This project proposes an objective machine-learning approach to detect ADHD in children using discriminative patterns extracted from EEG brain signals. **Input:** EEG signal **Output:** Binary classification with probability **Reference:** https://arxiv.org/abs/2504.04664